

Lucas Hirsch

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Education

University of Edinburgh , Machine Learning, Engineering, AI • PhD in Machine Learning	Edinburgh, UK 2025 – present
University of Buenos Aires , Electronic Engineering, Signal Processing • My thesis involved studying the application of compressed sensing and sparsity techniques to photoacoustic tomography image reconstruction.	Buenos Aires, Arg. 2014 – 2020

Experience

University of Edinburgh , PhD Student PhD student in Machine Learning • Researching machine learning methods for data-driven computational sensing and imaging in limited data . • Emphasis on satellite and SAR imagery.	Edinburgh, UK 2025 – present 1 year
Tenaris, Applied Physics Group , R&D Engineer R&D at the Applied Physics Group developing optical non-destructive testing systems • Researched non-destructive testing optical metrology systems and techniques for micrometer defect detection in pipe threads. • Developed multiple signal processing algorithms for filtering, defect detection and measuring various thread parameters in industrial settings .	Buenos Aires, Arg. 2021 – 2025 4 years
University of Buenos Aires , Research Assistant Researching the photoacoustic imaging inverse problem. • Developed algorithms for reconstructing images from (photo)acoustic pressure measurements. • Worked on applying compressed sensing and sparse reconstruction techniques to reduce the complexity of a photoacoustic tomography system.	Buenos Aires, Arg. 2021 – 2021 1 year

Teaching Experience

University of Edinburgh , Digital Signal Processing Laboratory, Machine Learning in Signal Processing, Signals and Communication Systems 3	Edinburgh, UK 2026 – present
University of Buenos Aires , Semiconductor Devices	Buenos Aires, Arg. 2017 – 2025
University of Buenos Aires , Programming and Algorithms II	Buenos Aires, Arg. 2018 – 2021

Awards

3rd Place — MAVIC-C 2026 Placed 3rd overall and 1st in out-of-distribution detection in the 5th Multi-modal Aerial View Imagery Challenge, out of 63 teams. Paper available here Royal Swedish Academy of Sciences	2026
Leaders of the Future 2024 Chosen by Tenaris as one of thirty young professionals to participate in the 2024 edition of " Leaders of the Future ". Royal Swedish Academy of Sciences	2024

16th International Microelectronics Olympiad Finalist

2021

One of three Latin American students chosen to represent the region on the 16th edition of the **International Microelectronics Olympiads**.

Royal Swedish Academy of Sciences

15th International Microelectronics Olympiad Finalist

2020

One of three Latin American students chosen to represent the region on the 15th edition of the **International Microelectronics Olympiads**.

German Physical Society

Languages

Spanish

Native speaker

English

Bilingual

German

Basic